
Mac-Diff

Conditional diffusion with locality-aware modal alignment for generating diverse protein conformational ensembles

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01

Motivation

Proteins are ensembles, and predicting one structure is not enough

Proteins function by moving between states

- Many proteins **interconvert** between conformational states of different free energy
- Function often lives in those transitions: **ligand binding, catalysis, allostery**, molecular evolution
- *Running example: adenylate kinase (AdK)* has a rigid CORE plus two flexible domains, **LID** and **NMP**, that open and close during $\text{ATP} \rightarrow \text{ADP}$ catalysis
- Drug discovery cares about the *non-dominant* states too: cryptic pockets, allosteric sites, transient conformations

The target is a **distribution** $p(\mathbf{x} \mid s)$ over 3D structures $\mathbf{x}$ for a sequence s : which states exist, and how populated they are.

How do we get ensembles today?

Experiment

- Typically one or a **few discrete, static** structures per protein
- Cost limits how many states are resolved

Molecular dynamics

- Physically grounded, **coherent trajectories**
- Folding needs **μs – ms** timescales, which is very expensive
- DESRES reference runs used the special-purpose **Anton** supercomputer
- WW domain in this paper: 100 μs $\approx$ **222 GPU-days** (A100)

Structure prediction

- **AlphaFold2**: accurate structure from sequence + MSA
- Returns an individual, arguably **most probable** structure
- Default inference does not recover the heterogeneity seen in MD / experiments

Goal: MD-like ensembles at structure-prediction cost, **for proteins never seen in training.**

This paper in one slide

Task — given a sequence s , draw samples $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)} \sim p_{\theta}(\mathbf{x} \mid s)$ that match the equilibrium ensemble. We need both **fidelity** and **diversity**.

Several baselines are faithful but collapse onto the folded state, and others are diverse but unfaithful.

Mac-Diff — score-based diffusion on a 2D inter-residue **geometry tensor**, conditioned on **ESM-2** through **LAMA-attention** (*locality-aware modal alignment*)

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Related work

AlphaFold2 perturbations · unconditional generators · conditional generators

Three routes to conformational diversity

A · Perturb AlphaFold2

- Make a deterministic predictor emit several structures
- **MSA** subsampling, residue replacement, clustering
- Dropout at inference, state-specific templates
- AF2 outputs as seeds for enhanced sampling
- Binding partners (AF2Complex)

B · Unconditional generators

- Learn $p(\mathbf{x})$ over structures
- Early: **GANs / VAEs** (idpGAN; VAE latent sampling)
- **FoldingDiff**: diffusion on backbone angles
- **Str2Str**: PDB-trained score model; perturb-and-denoise a start structure

C · Conditional generators

- Learn $p(\mathbf{x} \mid s)$ from a sequence representation
- **EigenFold · ConfDiff · DiG · BioEmu**
- **AlphaFlow**: flow matching on AF2
- **IdpSAM**: latent diffusion for IDPs
- **Mac-Diff** sits here, with a **PLM** condition

AlphaFold2 in one slide: why the MSA matters

- **Co-evolution**: residues in contact tend to mutate together, so a deep **multiple sequence alignment (MSA)** encodes contacts
- The **Evoformer** jointly refines an MSA representation and a residue-**pair** representation, including *triangle multiplicative updates* (Mac-Diff reuses these later)
- The structure module (invariant point attention) turns pair features into **one** 3D structure
- The MSA pools signals from the **whole family**, but default inference returns one state, usually the dominant one
- Chakravarty et al. (2024): AF2 predictions of fold-switched states are largely driven by **structure memorization**

The MSA is the "knob" that most ensemble tricks turn, and the source of bias the Mac-Diff authors want to avoid.

Getting ensembles out of AlphaFold2

Method	Knob	Idea
MSA subsampling — del Alamo et al. 2022; Monteiro da Silva et al. 2024	MSA depth	Shallow, random MSAs weaken the dominant co-evolutionary signal, so alternative states appear; the 2024 work also estimates relative populations
SPEACH_AF — Stein & Mchaourab 2022	MSA columns	In-silico alanine substitutions across MSA columns (the paper's "residue replacement")
AF-Cluster — Wayment-Steele et al. 2024	MSA composition	Cluster the MSA (DBSCAN) and run AF2 per cluster. Different sub-families give different states (e.g. fold-switchers)
Dropout at inference	network	Keep dropout on at test time and sample with many seeds
State-specific templates — Heo & Feig 2022	templates	Active/inactive templates steer GPCR predictions
AF2-RAVE — Vani et al. 2023	downstream	Reduced-MSA AF2 structures seed enhanced MD sampling and Boltzmann ranking
AF2Complex — Gao et al. 2022	partners	States that appear only in the presence of interaction partners

Shared limits: needs a **deep MSA**, sampling is heuristic (no calibrated populations), and default inference still under-samples heterogeneity.

Unconditional generative models

- **Early deep generative models:** GANs and VAEs over conformations (idpGAN, Janson et al. 2023; VAE latent sampling, Mansoor et al. 2024; Tian et al. 2021)
- **Diffusion models** took over (Sohl-Dickstein 2015 → score SDEs, Song et al. 2021) because they give realistic samples from the training distribution
- **FoldingDiff** (Wu et al. 2024): denoising diffusion over backbone **internal angles**, unconditional
- **Str2Str** (Lu et al., ICLR 2024): score model trained on the **PDB only**. It noises an input structure to an intermediate time and denoises it, giving *zero-shot* conformation sampling without MD data
 - two samplers, **SDE** and **probability-flow ODE (PF)**
- Representation lineage used by Mac-Diff: **trRosetta** (2020) predicts inter-residue distances and orientations. **ProteinSGM** (Lee et al. 2023) runs score-based diffusion on those 2D maps and then Rosetta folding (for design)

These models learn "what proteins look like"; the open question is how to inject **sequence-specific** information.

Conditional generative models

- **EigenFold** (Jing et al. 2023): *harmonic* diffusion over C α eigenmodes, conditioned on **OmegaFold** embeddings
- **ConfDiff** (Wang et al. 2024): force-guided **SE(3)** diffusion, conditioned on **ESMFold** embeddings
- **DiG** (Zheng et al. 2024): Distributional Graphormer on PDB + MD, conditioned on **AF2 Evoformer** embeddings
- **BioEmu** (Lewis et al. 2025): IPA-based diffusion, trained on tens of millions of structures + > 200 ms MD, conditioned on **AF2 Evoformer** embeddings
- **AlphaFlow-MD** (Jing et al. 2024): AF2 weights fine-tuned with **flow matching** on ATLAS MD
- **IdpSAM** (Janson & Feig 2024): latent diffusion transformer for **intrinsically disordered** proteins

Common pattern: the geometry (frames, C α coordinates), the denoiser (IPA) **and** the conditioning embedding all come from **structure-prediction models** trained to output one structure.

The gap Mac-Diff targets

1 · What is the condition?

- Structure-predictor embeddings are trained toward **one** structure and may be biased to the **dominant** state
- Mac-Diff uses a **protein language model** instead: ESM-2, trained with masked-token prediction on massive sequence sets, which captures evolutionary, structural and functional signal
- **Alignment-free**, with no MSA search

2 · How is it injected?

- Text-to-image diffusion uses **cross-attention**, where each pixel can attend to every word: a loose, global, purely data-driven alignment
- For sequence $\rightarrow$ structure, residue i 's own coordinates are trivial. What matters is **which sequence positions are its 3D neighbours**
- Mac-Diff's **LAMA-attention** biases each residue's attention toward residues it likely contacts

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Method

Representation · diffusion · denoiser · LAMA-attention · folding

Mac-Diff at a glance

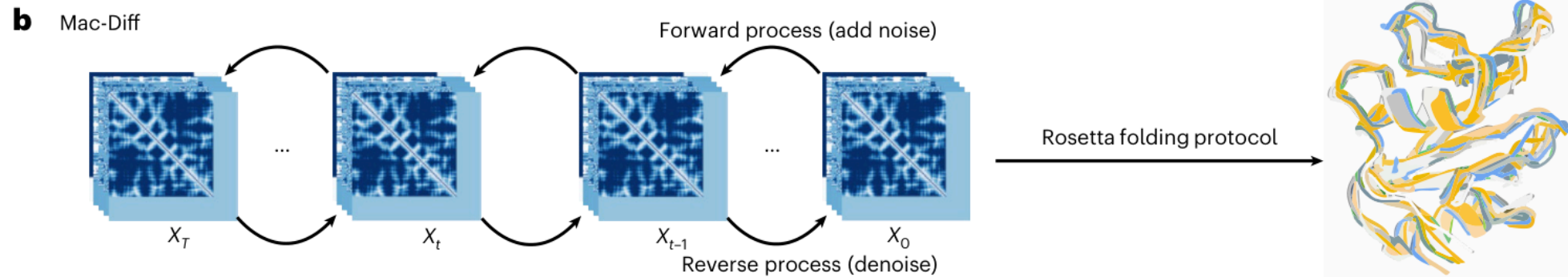


Fig. 1b (top). Diffusion runs on 2D geometry maps, and Rosetta converts the final map into a 3D structure

1. **Sequence** $s \rightarrow$ ESM-2 (650M): residue embeddings $s_i \in \mathbb{R}^{1280}$ + predicted **contact map** C
2. **Noise / denoise** a geometry tensor $\mathbf{X} \in \mathbb{R}^{L \times L \times 5}$ with a conditional score network $s_\theta(\mathbf{X}_t, t, s)$
3. **Sample** with DDIM (50 steps): 1,000 tensors per protein in Task 1, 100 in Task 2
4. **Fold** each tensor into an all-atom structure with **Rosetta**, using trRosetta-style restraints

Representation: a residue-pair "image"

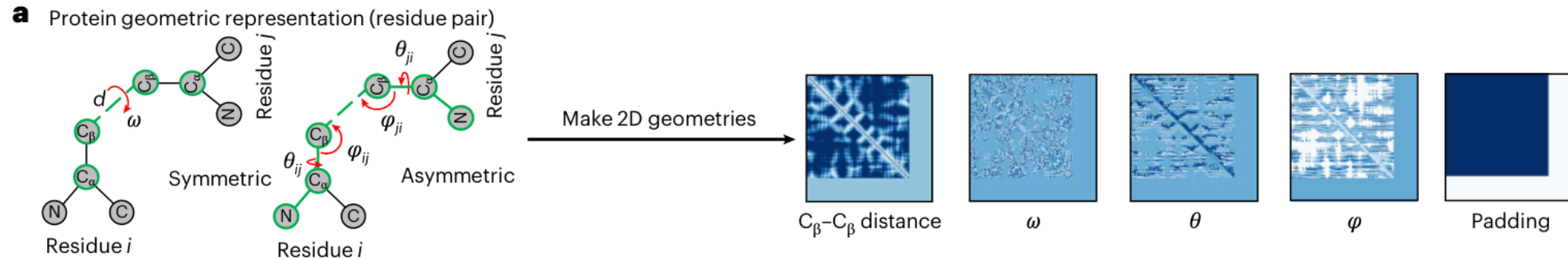


Fig. 1a. Five channels per residue pair (i, j)

- **C_β - C_β distance** d_{ij} · dihedral ω (symmetric, $\omega_{ij} = \omega_{ji}$) · dihedral θ and planar angle φ (asymmetric, $\theta_{ij} \neq \theta_{ji}$) · a **padding** channel marking the true length L
- These are exactly **trRosetta**'s inter-residue geometries; ProteinSGM already diffused over such maps
- ✓ **invariant** to global rotation and translation, so no SE(3) machinery is needed and the tensor can be treated as a multi-channel image
- ✗ over-complete: $O(L^2)$ numbers for $3L$ backbone positions. An arbitrary tensor need not be realizable in 3D, so a **folding step** is required

Score-based conditional diffusion (VP-SDE)

Forward: noise the geometry tensor toward $\mathcal{N}(0, I)$

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}dt + \sqrt{\beta(t)}d\mathbf{w}, \quad \beta_{\min} = 0.1, \beta_{\max} = 20$$

Reverse: requires the score $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$

$$d\mathbf{x} = [f(\mathbf{x}, t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})] dt + g(t) d\bar{\mathbf{w}}$$

Training: conditional denoising score matching (Song et al. 2021)

$$\theta^* = \arg \min_{\theta} \mathbb{E}_t \left\{ \lambda(t) \mathbb{E}_{\mathbf{x}(0)} \mathbb{E}_{\mathbf{x}(t)|\mathbf{x}(0)} \left\| s_{\theta}(\mathbf{x}(t), t, \text{seq}) - \nabla_{\mathbf{x}(t)} \log p_t(\mathbf{x}(t) | \mathbf{x}(0)) \right\|_2^2 \right\}$$

- $t \sim \mathcal{U}(0, T)$, $N = 2,000$ discretization steps in training, and a **DDIM sampler with 50 steps** at inference

The denoising network: a 2D U-Net with attention

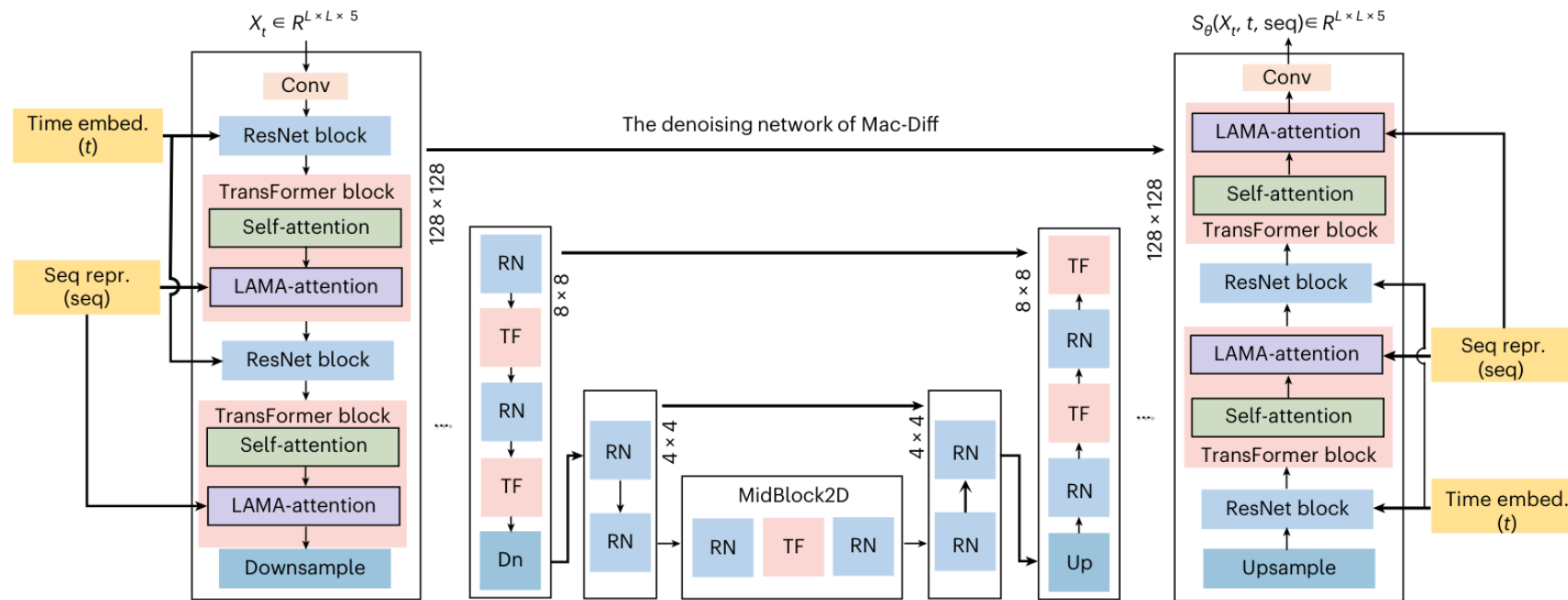


Fig. 1b (bottom). RN = ResNet block · TF = Transformer block (self-attention + LAMA-attention)

- 3×3 conv lifts 5 → **128 channels**; **5 down / 5 up stages** with skip connections ($128 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 8$)
- Per stage, twice: **ResNet block** (two 3×3 convs; time embedding added after the first) → **Transformer block** (self-attention over the L^2 pair features, then **LAMA-attention** injects the sequence)
- Down: 2×2 mean-pool, stride 2 · up: 2×2 nearest-neighbour interpolation

Why not plain cross-attention?

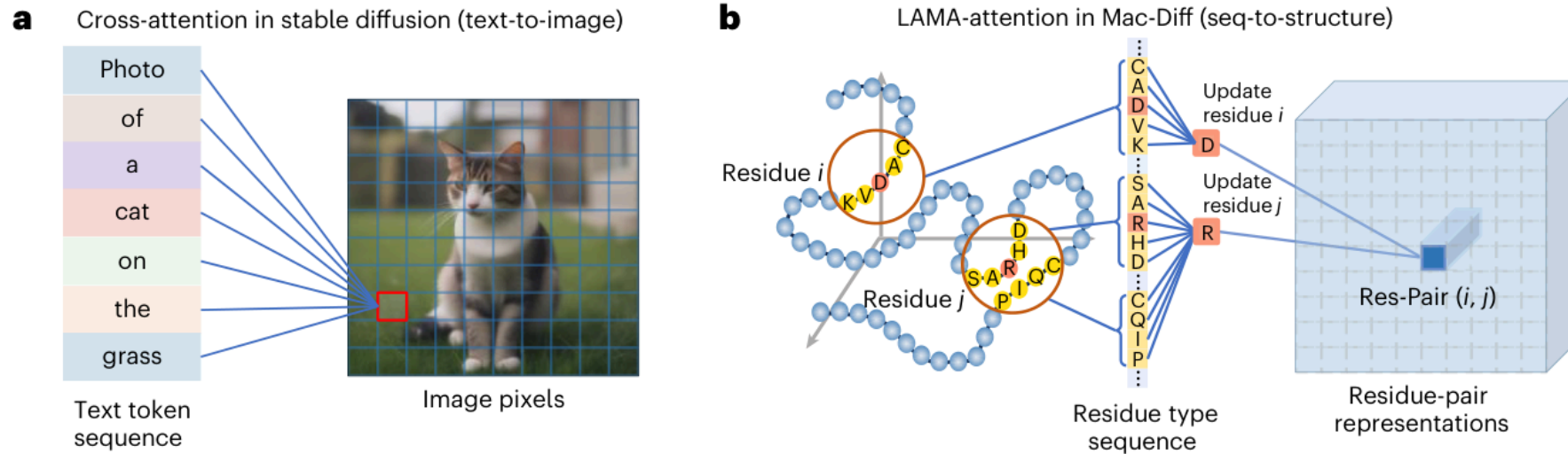


Fig. 2. (a) Stable Diffusion cross-attention; (b) LAMA-attention

- **Text-to-image:** a pixel queries **all** tokens, and the alignment is unstructured and learned purely from data
- **Sequence-to-structure:** pair (i, j) should be informed by residues i, j and their likely **spatial neighbours**, which can be far apart in the sequence
- LAMA "narrows the attention field from the whole sequence to only a small fraction of useful residues"

LAMA-attention (1): a relational matrix over residues

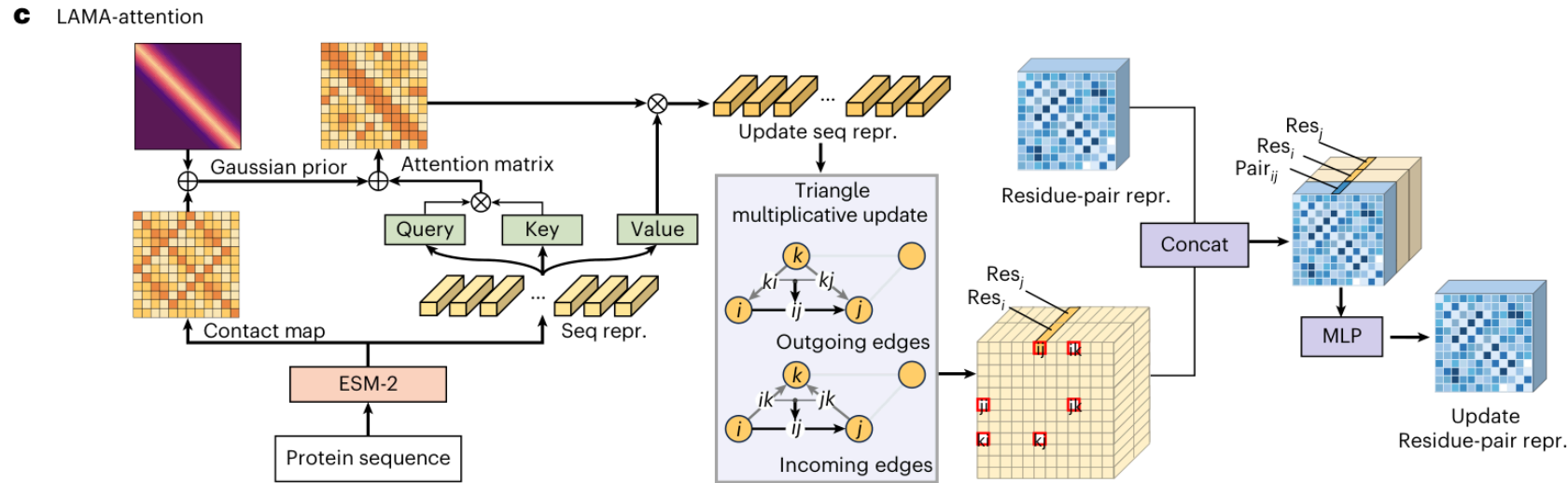


Fig. 1c. Gaussian prior + ESM-2 contact map + learned attention → updated residue features → triangle update → fused with pair features

$$W_{ij} = \underbrace{\lambda \frac{\exp(q_i^\top k_j / \sqrt{f_d})}{\sum_{j'} \exp(q_i^\top k_{j'} / \sqrt{f_d})}}_{\text{learned attention}} + (1 - \lambda) \left(\alpha \underbrace{C_{ij}}_{\text{ESM-2 contacts}} + \beta \underbrace{e^{-|i-j|^2 / 2h^2}}_{\text{chain-local prior}} \right)$$

$q_i = W_q s_i$, $k_i = W_k s_i$, $v_i = W_v s_i$ from ESM-2 embeddings $s_i \in \mathbb{R}^{1280}$. each row of W sums to 1

LAMA-attention (2): three sources of "who talks to whom"

Term	What it encodes	Reach	Setting
Learned attention $\text{softmax}(q_i^\top k_j)$	fine-grained, context-dependent interactions from ESM-2 features (biochemical identity, co-variation)	any pair	weight λ learned , initialized to 3×10^{-4}
ESM-2 contact map C_{ij}	logistic regression over ESM-2 attention heads, weakly supervised on ~a dozen structures. A coarse global prior on contacts	long-range	$\alpha = 0.3$
Gaussian band $e^{- i-j ^2/2h^2}$	residues adjacent in the chain are adjacent in space. A simple physical regularizer	local	$\beta = 0.7, h = 2$

LAMA-attention (3): from residue features to pair features

1. **Attend**: update residue features with the relational matrix, $\tilde{V} = WV$
2. **Pair up** (outer concatenation, Eq. 17): $z_{ij} = \text{Concat}[(WV)_i, (WV)_j]$
3. **Triangle update** borrowed from AlphaFold2: $\tilde{z}_{ij} = \text{TriMul}(z_{ij})$, reasoning over residue triplets $i-j-k$
4. **Fuse** with the noisy geometry features a_{ij} (output of self-attention):

$$\tilde{a}_{ij} = \text{Linear}\left(\text{GELU}\left(\text{Linear}(\text{Concat}[a_{ij}, \tilde{z}_{ij}])\right)\right)$$

and pass the result to the next down/up-sampling layer

q, k, v all come from the **sequence**; the noisy structure enters only at the fusion step. The "alignment" is by index (pair $ij \leftrightarrow$ residues i, j), and W supplies the locality. The sequence branch does not depend on $\mathbf{x}_t$, so in principle it could be computed once per protein.

From 2D geometry to 3D atoms: Rosetta

Task 1 · ensembles (trRosetta protocol)

- **Soften the predictions:** each predicted distance and angle becomes a smooth probability instead of a single exact value, so small errors are tolerated
- **Turn them into an energy:** geometries the model finds likely get low energy, unlikely ones get high energy
- **Fold by minimizing it:** build a coarse backbone that best satisfies all the pair restraints, then refine it with full atoms

Task 2 · alternative states (ProteinSGM-style)

- **Looser restraints:** each prediction is a target with a wide tolerance, because these proteins move more
- **Constrain only nearby pairs:** far-apart residues are left free, so domains can shift
- **Several tries, keep the best:** fold several times and keep the lowest-energy structure

The predicted 2D map is not guaranteed to be a consistent 3D structure; Rosetta finds the structure that best agrees with it. It is a separate, non-differentiable step, and it is **set up differently for each task**.

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Training

Learn what folded proteins look like from the PDB, then learn how they move from MD

Training data

PDB: pre-training

- Train on entries $\leq$ **1 May 2022**; validate on 2 May 2022 – 2 Jan 2023
- Keep **all chains**, not just the first, to expose conformational heterogeneity
- Cropped segments
 - **619,045** $\times$ 40–128 residues $\rightarrow$ Mac-Diff-128
 - **579,476** $\times$ 80–256 residues $\rightarrow$ Mac-Diff-256

MD: fine-tuning on 1,674 trajectories

- **GPCRmd**: 371 GPCRs (255–501 residues) in lipid bilayers, CHARMM36m, 3×500 ns at 310 K
- **ATLAS**: 1,390 non-membrane chains, CHARMM36m / GROMACS, 3×100 ns
- **Frame selection**: TTClust $\rightarrow$ 100 clusters per trajectory $\times$ 10 random frames, giving 1,000 frames per protein (≈ 1.67 M frames per epoch)
- Split: **1,674 train / 80 validation** trajectories

Redundancy control and fairness

- Any PDB or MD protein with **> 40 % sequence identity** (MMseqs2) to **any** test protein is removed, in **both** stages (similar to ESMFold's filtering)
- Baselines are **released checkpoints** trained on their original data **without** this filter (retraining would be prohibitively costly)
 - e.g. **8 of 40** Cfold40 proteins have **> 90 %** identity to BioEmu's training data
- Test trajectories run 100 μ s – ms, versus 100–500 ns in training, so the benchmark also tests **temporal generalization**

Conservative for Mac-Diff on leakage, but **not a controlled architecture comparison**: data, scale and embeddings all differ.

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Evaluation

Two tasks with two kinds of ground truth

Two tasks

Task 1 · recover MD distributions

- **12 fast-folding proteins** (Lindorff-Larsen et al. 2011; DESRES/Anton): Chignolin (10 aa) to Lambda (80 aa), $\geq$ **10 folding–unfolding events** each, $\sim 0.1\text{--}3$ ms of simulation
- **BPTI** (58 aa): a 1 ms trajectory with 5 known substates (Shaw et al. 2010)
- Reference: 1,000 frames at fixed stride ($C\alpha$). **1,000 samples** per method, 3 runs
- Baselines: AlphaFlow-MD, EigenFold, IdpSAM, Str2Str (PF / SDE), BioEmu

Task 2 · recover experimental alternative states

- **AdK** (*E. coli*): open **4AKE** / closed **1AKE**
- **Cfold40**: 40 of 245 Cfold proteins (Bryant & Noé 2024) with < 256 residues, < 8 consecutive missing residues, and two states with TM < 0.8 (hinges, rearrangements, fold switches)
- **100 samples** per method; MSAs from the ColabFold server
- Baselines: AF2, AF2 + dropout, MSA subsampling, AF-Cluster, AlphaFlow-MD, DiG, Boltz-2, BioEmu

Metrics: distributions (Task 1)

- **JS-PwD**: histogram (50 bins, $\varepsilon = 10^{-6}$) of C α –C α distances with sequence separation $\geq 4 \rightarrow$ **global shape**
- **JS-Rg**: histogram of radius of gyration $\rightarrow$ **compactness**
- **JS-TIC**: project pairwise distances onto the MD's top-2 **TICA** components (Deeptime), 50 bins per dimension, averaged $\rightarrow$ **slow, folding-relevant motions**
- **Diversity** $= 1 - \frac{1}{N_{\text{pairs}}} \sum_{i < j} \text{IDDT}(s_i, s_j)$ · **Fidelity** $= 1 - \text{JS-PwD}$
- **EMD**: optimal transport between the generated and MD ensembles, with cost $1 - \text{TM-score}$ (global) or $1 - \text{IDDT}$ (local)
- **RMSE**_{cmap} on log contact probabilities (C α < 10 Å) · **RMSF** correlation · **TICA recall** (50 $\times$ 50 grid) · metastable-state recovery (RMSD < 2 Å folded, < 3 Å others)

Metrics: alternative states (Task 2)

- **Best match:** highest TM-score / lowest RMSD between any sample and each reference state
 - success thresholds **TM** ≥ 0.8 or **RMSD** ≤ 3 Å
- **Coverage:** both / one / neither state recovered; worst case = min TM (max RMSD) over the two states
- **Diversity:** TM_{var} = mean pairwise TM within the ensemble; **MAE diversity**
 $= |TM_{\text{var}} - TM(\text{conf}_1, \text{conf}_2)|$
- **Flexibility:** per-residue RMSF of the ensemble vs. the references $\rightarrow$ MAE and Pearson correlation
- **Statistics:** two-sided Wilcoxon signed-rank tests on the top-3 predictions $\times$ 2 states $\times$ 40 proteins

06

Results

Fast-folders · BPTI · ablations · AdK & Cfold40 · speed

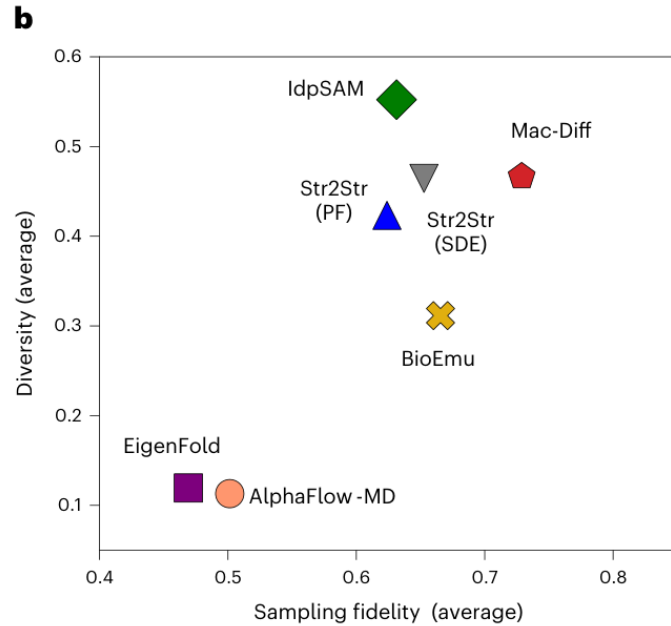
Task 1 · Fast-folding proteins: distribution match

Mean over 12 proteins ↓	AlphaFlow-MD	EigenFold	IdpSAM	Str2Str (PF)	Str2Str (SDE)	BioEmu	Mac-Diff
JS-PwD	0.498	0.532	0.355	0.377	0.353	0.334	0.276
JS-Rg	0.529	0.654	0.433	0.454	0.366	0.354	0.275
JS-TIC	0.528	0.497	0.449	0.404	0.400	0.356	0.339

- vs. the best baseline (BioEmu): **−18 % / −22 % / −5 %**. Best per protein on 6 / 7 / 5 of 12 (Supp. Tables 3–5)
- Wilcoxon (n = 12, 3 runs each): significant vs. **5 of 6** baselines on every metric. **Against BioEmu, only JS-Rg** is significant (p = 0.034; JS-PwD p = 0.23, JS-TIC p = 0.91)
- **Chignolin** (10 aa, mostly loop): every method does poorly

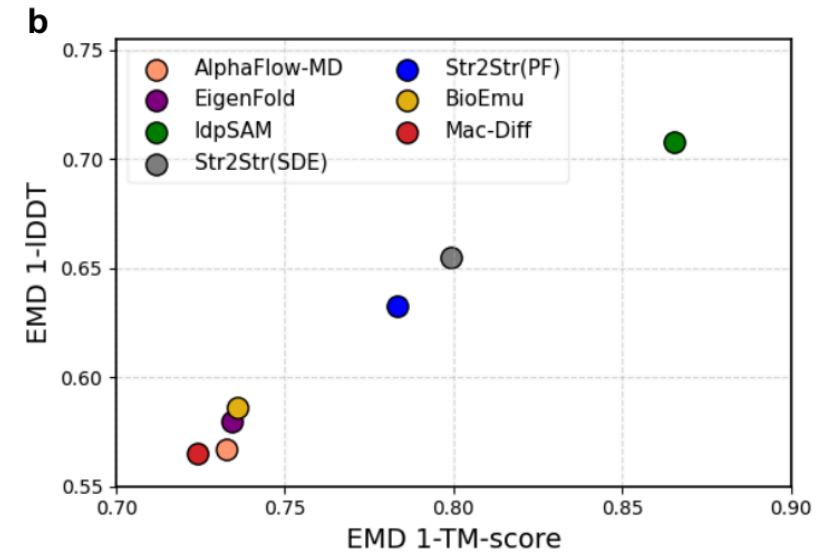
Fidelity *and* diversity

Diversity vs. fidelity (Fig. 3b)



mean over 12 proteins · top-right is better

Ensemble distance (Supp. Fig. 1b)

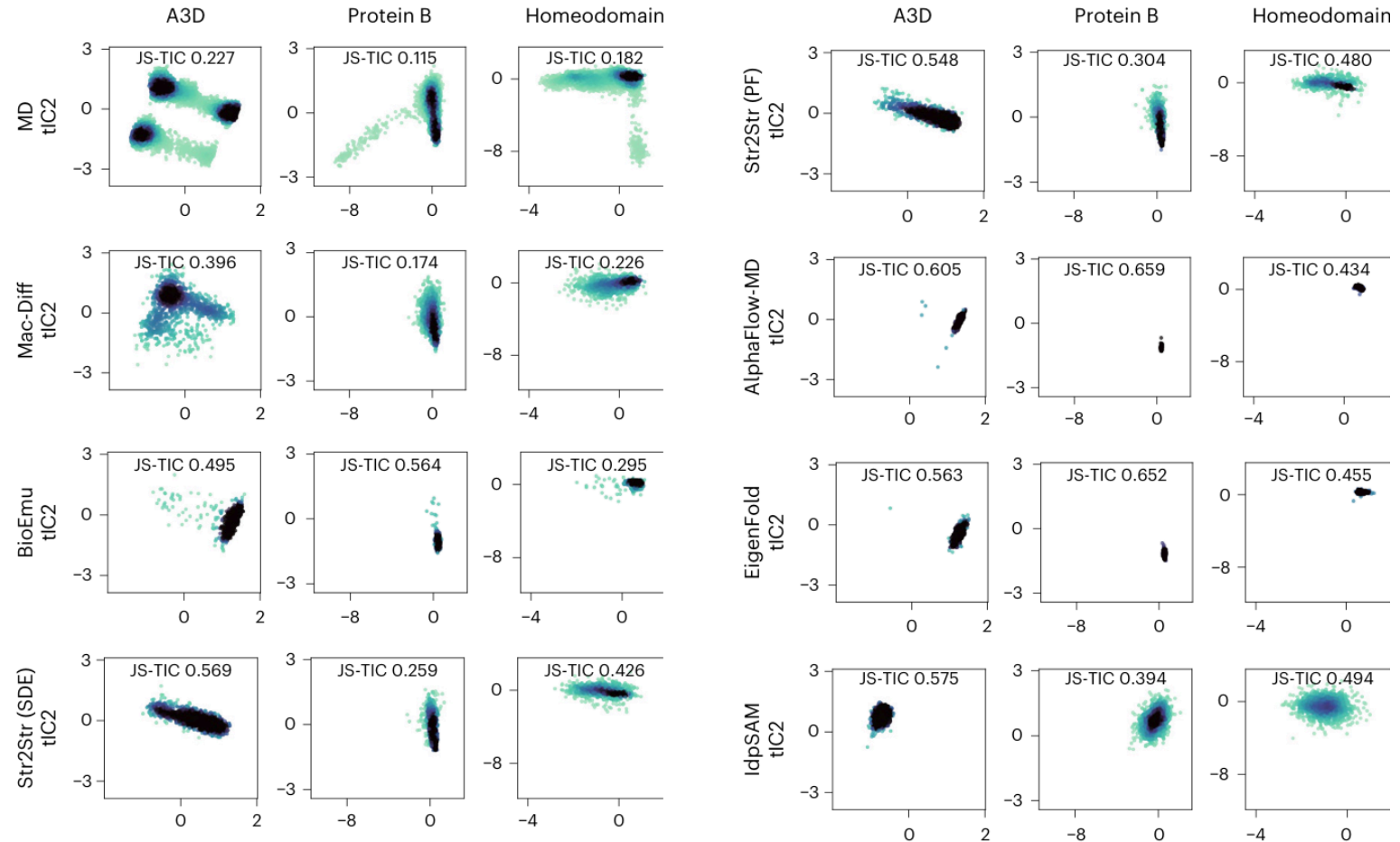


EMD with 1 - TM (x) and 1 - lDDT (y) costs · lower-left is better

Mac-Diff has the highest fidelity (≈ 0.72) at high diversity (≈ 0.47). AlphaFlow-MD and EigenFold **collapse** onto the folded state. On EMD, Mac-Diff is lowest, but **AlphaFlow-MD, EigenFold and BioEmu are close.**

TICA landscapes

- One dot per sample, projected on the MD's top-2 TICs
- JS-TIC vs. the best baseline
 - α 3D **0.396** vs 0.495 (−20 %)
 - Protein B **0.174** vs 0.259 (−33 %)
 - Homeodomain **0.226** vs 0.295 (−23 %)
- MD self-consistency floor: 0.227 / 0.115 / 0.182
- TICA recall, 12 proteins: **Mac-Diff 0.403** $\approx$ Str2Str-SDE 0.401 (best two)



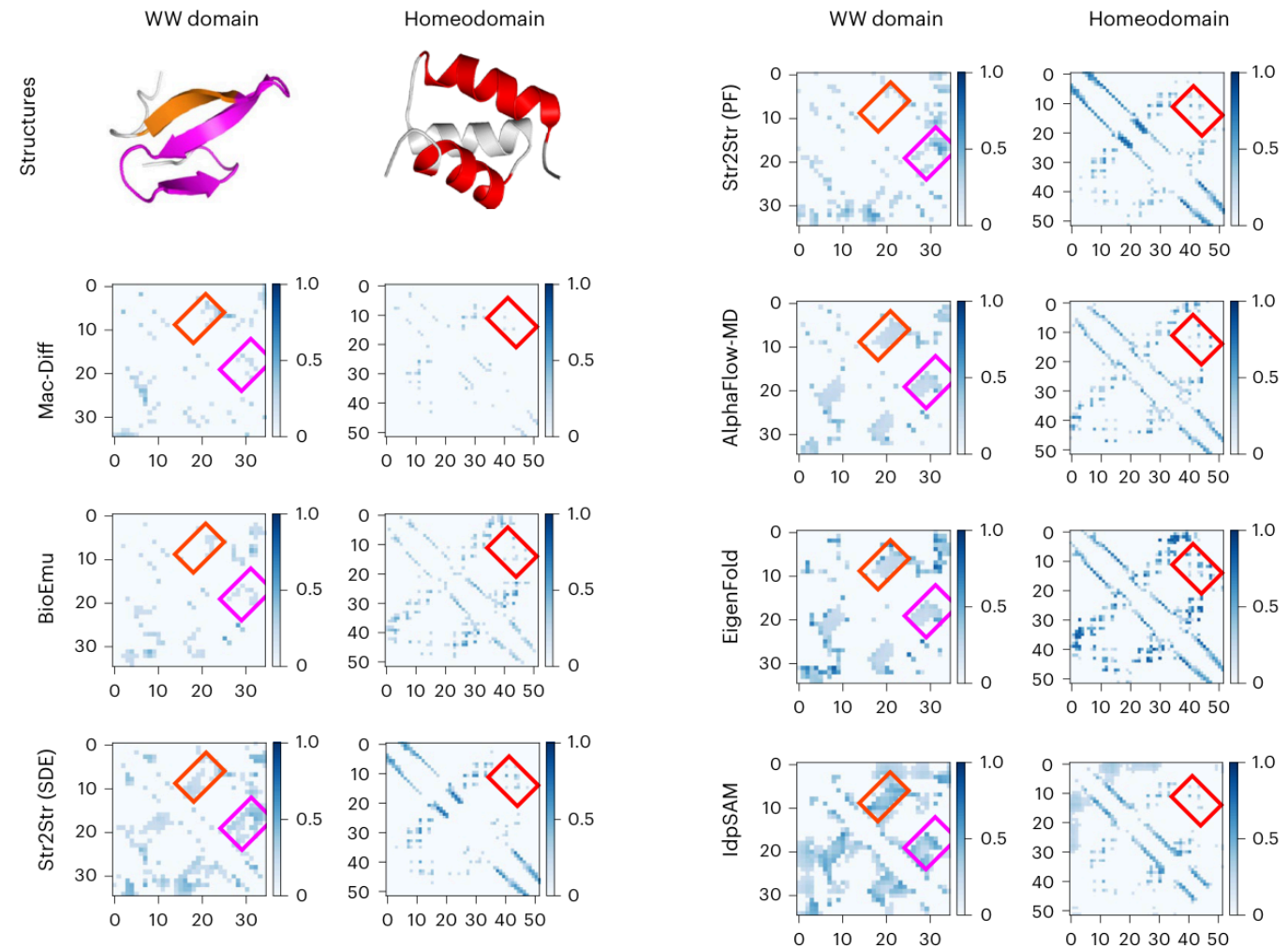
Metastable states, flexibility and contacts

12 fast-folders	AlphaFlow-MD	EigenFold	IdpSAM	Str2Str (PF)	Str2Str (SDE)	BioEmu	Mac-Diff
Folded recovered (RMSD < 2 Å)	11	11	1	9	8	11	11
Loosely packed (< 3 Å)	0	0	6	9	9	4	10
Unfolded (< 3 Å)	0	0	2	0	0	0	2
RMSF correlation (median r) ↑	0.68	0.56	0.73	0.83	0.86	0.91	0.89
RMSE contact map (mean) ↓	11.40	10.31	1.42	1.12	1.11	4.69	1.07

- States: TICA → 3 clusters per protein (folded / loosely packed / unfolded), following Zheng et al. (DiG)
- The largest gap is in **loosely packed** states (10 / 12 vs ≤ 9); **unfolded** states remain hard for everyone because of their high heterogeneity

Contact maps

- |generated – MD| contact probability;
lighter is better
- **WW domain** (3-stranded β -sheet) and **Homeodomain** (3 helices)
- AlphaFlow-MD, EigenFold: native contacts only, missing the unfolded but physically feasible patterns
- IdpSAM: IDP-like β patterns (magenta box)
- Str2Str: spurious weak $\beta 2 \leftrightarrow$ C-terminus contacts
- Mac-Diff: gets the helix 1 $\leftrightarrow$ helix 3 packing (red box)



BPTI: substates from a 1 ms simulation

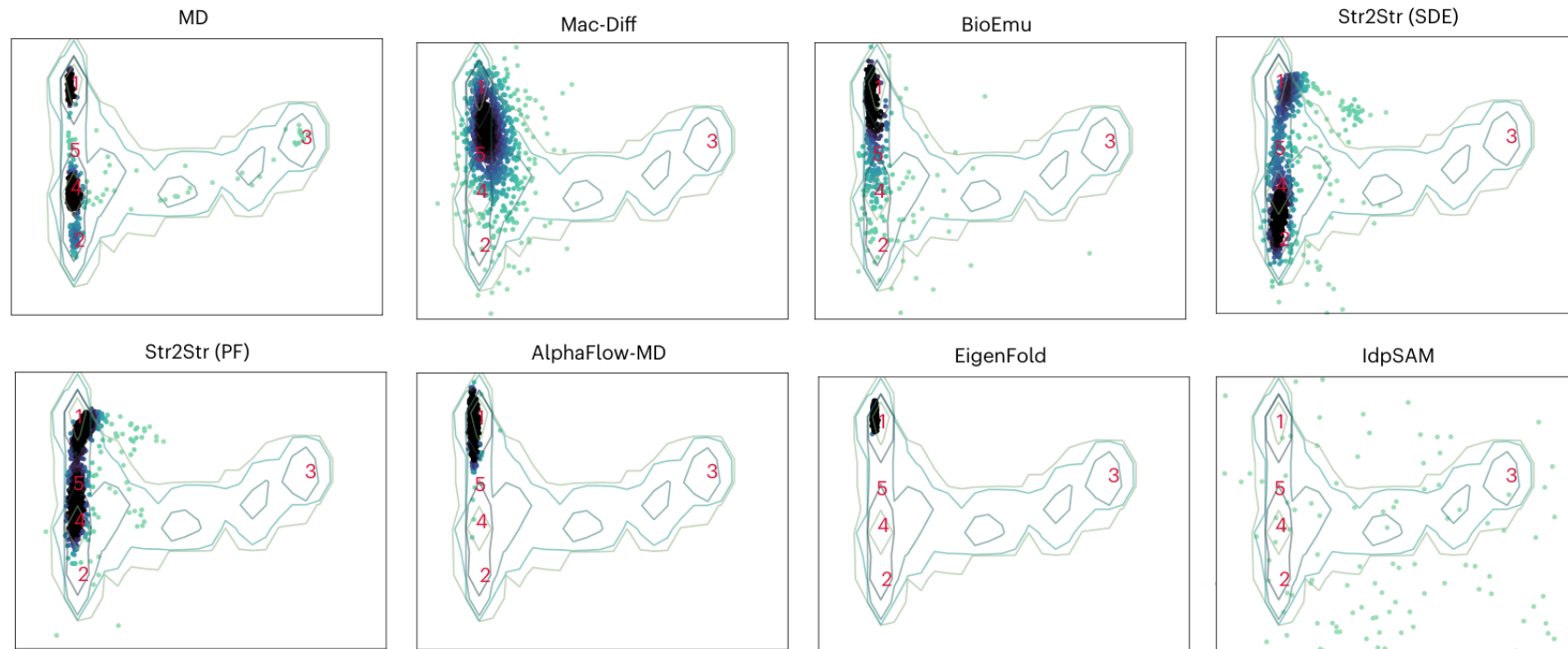


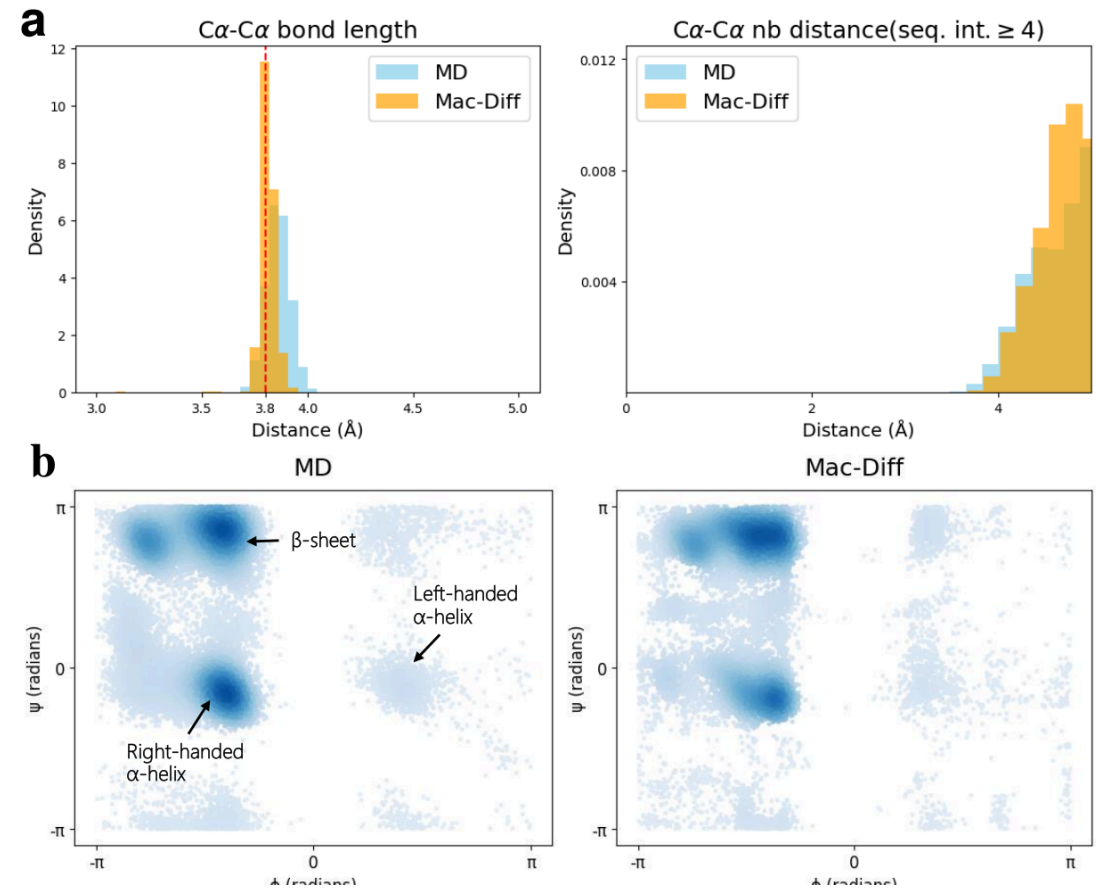
Fig. 5 (rearranged). Samples projected on the MD's top-2 TICs; red numbers mark the 5 known substates

- Best-match RMSD: lowest for clusters **3, 4, 5** (2.18, 1.26, 0.84 Å) and lowest mean, **1.33 Å**, but not significant vs. AlphaFlow-MD, EigenFold or BioEmu (Bonferroni)
- Distribution level: **BioEmu is better** on BPTI (JS-PwD 0.236 vs 0.317; JS-TIC 0.350 vs 0.494)

Physically valid?

- **C α –C α bond length**: mean **3.81 Å** (MD 3.85 Å; ideal 3.8 Å)
- **Closest non-bonded C α pair** (sequence separation ≥ 4): **3.49 Å** vs 3.48 Å in MD, so no self-crossings
- **Ramachandran**: right-handed α , β and left-handed α regions are populated as in MD

The 2D restraints are realizable, and Rosetta turns them into valid backbones.



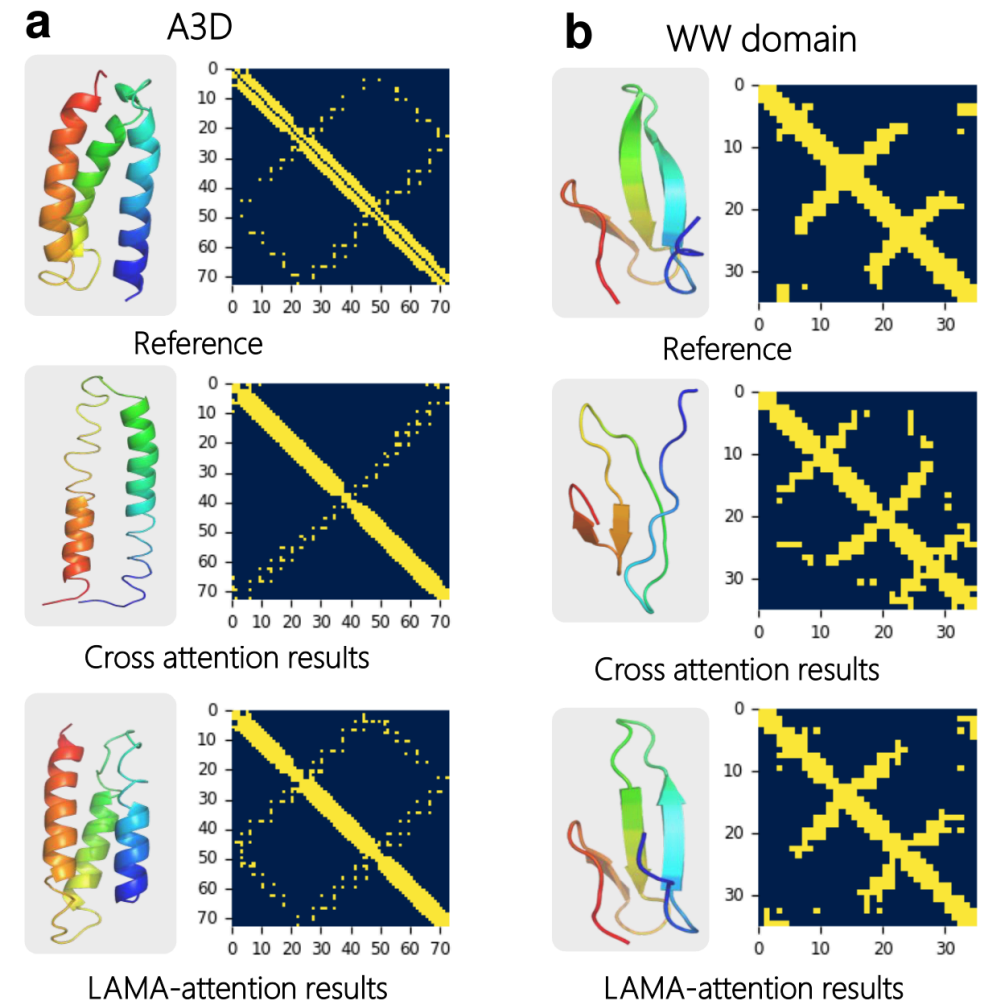
Cross-attention vs. LAMA

- Swap LAMA for **Stable Diffusion's cross-attention**; same network and MD data

α 3D: the cross-attention variant "exhibited a **completely incorrect contact pattern** when compared to the reference contact map", giving a **double helix** instead of three helices.

WW domain: contacts "were **perpendicular to the main diagonal** but failed to capture the correct positions", giving "four antiparallel β -sheets, rather than two antiparallel β 1 β 2 and β 2 β 3 sheets".

- Caveat: qualitative, one sample per model, no metrics



Ablation: what goes into the relational matrix

W built from	JS-PwD ↓	JS-Rg ↓	JS-TIC ↓
Gaussian prior only	0.471	0.659	0.472
Gaussian + contact map	0.352	0.558	0.409
Gaussian + residue embedding	0.345	0.439	0.399
Gaussian + wrong contact map + residue embedding	0.443	0.541	0.428
Gaussian + contact map + residue embedding (full)	0.306	0.352	0.363

- Tested variants: "(1) only the Gaussian prior, (2) + contact map, (3) + residue embedding, (4) all three, (5) a **wrong contact map** simulated by a totally randomized contact probability matrix"
- **+ contact map**: error down by up to **25 %** · **+ embeddings** on top: down by up to **37 %** (JS-Rg 0.558 → 0.352)
- **Wrong contacts**: **+45 % / +54 % / +18 %** vs. full, **worse than using no contact map**

Task 2 · Allosteric protein AdK: open and closed

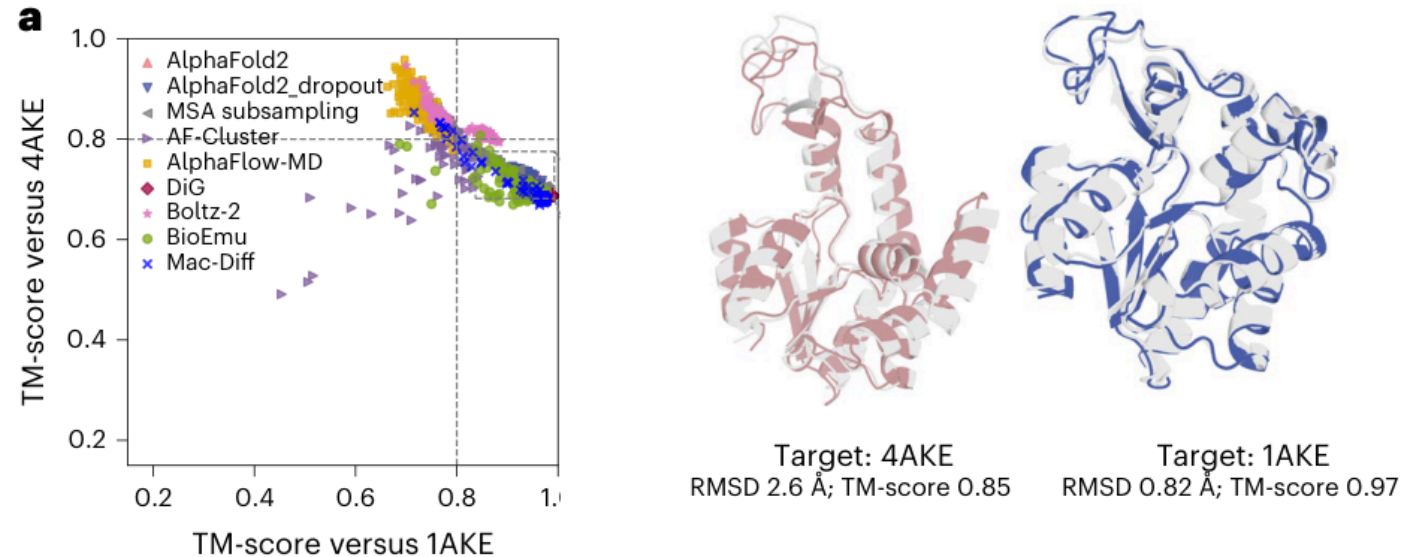


Fig. 6a. 100 samples per method; axes are TM-scores to closed (1AKE) and open (4AKE); best Mac-Diff samples overlaid on the targets

- Mac-Diff covers **both** states: best TM **0.85** to open (2.6 Å) and **0.97** to closed (0.82 Å), tied best with **MSA subsampling** (0.98 / 0.84)
- AF2, AF2 + dropout and DiG stay **closed**; AlphaFlow-MD and Boltz-2 lean **open**; BioEmu gets closed well, open less so

Task 2 · Cfold40 examples

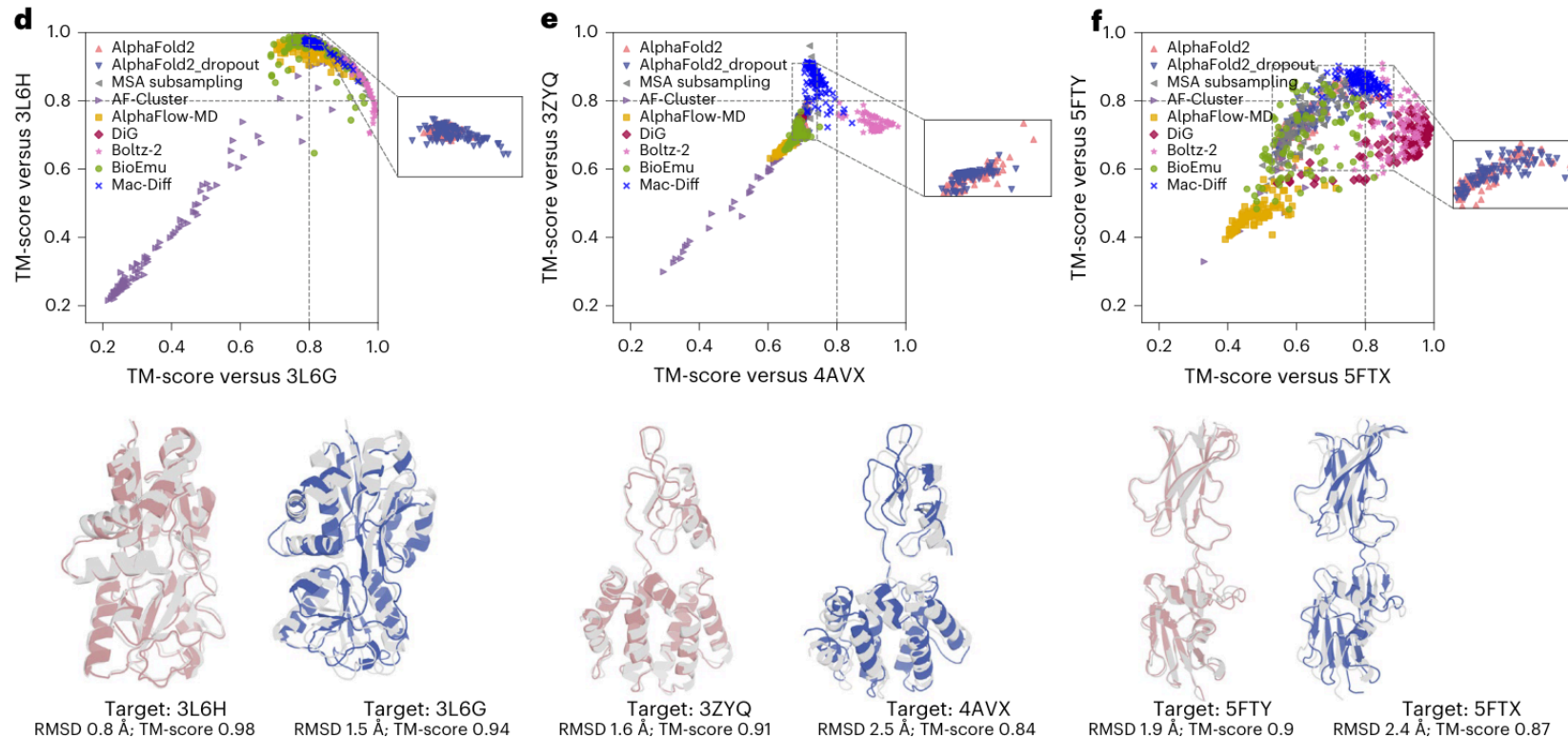


Fig. 6d-f. SBDs of OpuA · HGS-Hrs VHS/FYVE domains · surface-layer protein. Claret/blue = best Mac-Diff samples vs. the two targets

- Five showcased proteins (plus TAF1 and PDI, Fig. 6b–c): **TM > 0.8 for both states in all five**, RMSD < 3 Å in four; the best competitor manages 4 and 3

Task 2 · Cfold40 overall (40 proteins × 2 states)

Method	Mean TM ↑	Mean RMSD (Å) ↓	# TM ≥ 0.8 (/80)	# RMSD ≤ 3 Å (/80)	Both states, TM (%) ↑	Neither state, RMSD (%) ↓	Worst-case TM ↑
AlphaFold2	0.84	4.0	48	49	32.5	15.0	0.76
AF2 + dropout	0.85	3.9	51	47	35.0	12.5	0.77
MSA subsampling	0.86	3.2	57	53	47.5	7.5	0.79
AF-Cluster	0.83	3.6	53	48	47.5	22.5	0.78
AlphaFlow-MD	0.84	3.7	52	49	45.0	17.5	0.78
DiG	0.80	5.0	37	36	10.0	25.0	0.71
Boltz-2	0.84	4.0	47	46	32.5	17.5	0.76
BioEmu	0.87	2.9	60	55	60.0	15.0	0.82
Mac-Diff	0.85	3.0	56	55	50.0	5.0	0.79

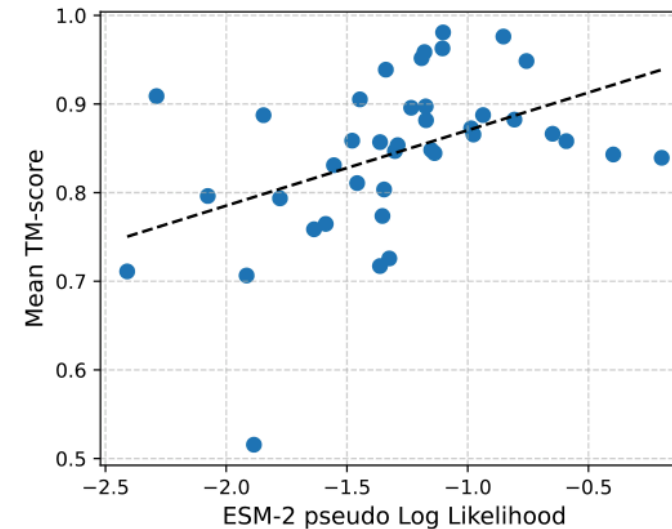
- Wilcoxon: **TM** is significant vs. MSA subsampling ($p = 0.048$) and DiG ($p = 0.001$) only. **RMSD** is significant vs. AF2, AF-Cluster, AlphaFlow-MD, DiG and Boltz-2. **Not significant vs. BioEmu** on either
- The **lowest "neither state" rate** (5 %) makes Mac-Diff the most reliable at finding at least one state

Diversity, flexibility, and why BioEmu stays ahead

Diversity & flexibility (Supp. Table 17)

Cfold40	MAE div. ↓	MAE flex. ↓	Flex. r ↑
BioEmu	0.122	3.35	0.57
AlphaFlow-MD	0.166	3.62	0.58
AF-Cluster	0.170	7.11	0.34
Mac-Diff	0.185	3.39	0.60
MSA subsampling	0.226	4.41	0.62
DiG	0.273	4.71	0.34

The embedding limits accuracy (Supp. Fig. 7)



ESM-2 pseudo-log-likelihood vs. mean TM · $r = 0.45$, $p = 0.004$

BioEmu has ~10× more training data, MSA-informed Evoformer embeddings, and no redundancy filtering (8 / 40 test proteins > 90 % identical to its training data).

Speed

vs. MD (WW domain, one A100)

Error vs. full MD (~1 ms)	MD 100 μ s	Mac-Diff
JS-PwD	0.399	0.289
JS-Rg	0.406	0.245
JS-TIC	0.438	0.441
Wall-clock	222 GPU-days	1.6 GPU-hours

Where the time goes (1,000 samples)

Stage	Runtime
Feature preparation (ESM-2)	1 s
Diffusion (VP-SDE, DDIM 50 steps)	~1.4 h
Rosetta folding (PyRosetta, parallel)	~0.2 h
Total	~1.6 h

$\approx 3,300\times$ faster than 100 μ s of explicit-solvent MD at comparable or better distributional error, but no kinetics, and slower than generators that emit Cartesian coordinates directly.

07

Discussion

Takeaways

1. **2D geometry diffusion + a PLM condition** is competitive with AF2-embedding-based generators, **without MSAs** and with $\sim 10\times$ less data than BioEmu
2. **How** the condition is injected matters. A chain-local prior + PLM contact prior + learned attention (LAMA) beats generic cross-attention, at least qualitatively, and **wrong priors hurt more than no priors**
3. Strongest on **distribution-level** metrics for fast-folders (Rg, loosely packed states, contact maps); **on par with, not better than, BioEmu** for alternative states
4. **$\sim 3,000\times$ faster than MD**, so large ensembles become affordable

Key references

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